

Durgesh B. Dongre

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EDUCATION

Year	Degree/Certificate	Institute	CPI/%
2024-Present	M.Tech. Electrical Engineering	Indian Institute of Technology, Kanpur	7.12/10
2018-2022	B.E. Electrical Engineering	Rashtrasant Tukadoji Maharaj Nagpur University	9.13/10
2018	Class XII (HSC)	MSBSHSE	70%
2016	Class X (SSC)	MSBSHSE	83.40%

MTECH THESIS

Federated Learning for Heterogeneous Commercial Image Domains (Ongoing)

Guide: Prof. Koteswar Rao Jerripothula , (Department of Electrical Engineering, IIT Kanpur).

- Developing communication-efficient **Fed-Cyclic** and **Fed-Star** federated learning algorithms to improve decentralized model aggregation while addressing **statistical heterogeneity across 8 commercial image domains**.

KEY PROJECTS

Synchronous 3-bit Up-Down Counter Design and Verification

Self-Initiated Project | Focus: Digital Logic Design & Verilog HDL | IIT Kanpur (Jan'26)

- Designed and implemented a **3-bit synchronous up-down counter** using **JK flip-flops in Verilog HDL**, deriving excitation logic via **state tables and Karnaugh maps**, and verified functionality using a **custom testbench in Xilinx Vivado**.

Hierarchical Design of 16:1 Multiplexer using Structural Modeling

Self-Initiated Project | Focus: Verilog HDL & Hierarchical RTL Design | IIT Kanpur (Dec'25)

- Designed a hierarchical **16:1 MUX** by instantiating **2:1 MUX** modules into **4:1 MUX** stages, utilizing a total of **fifteen 2:1 MUX** and verifying functionality via a comprehensive **Verilog testbench**.

CMOS Inverter Design and Analysis using GPDK 180nm Technology Node

Self-Initiated Project | Focus: Cadence Virtuoso (Nov'25)

- Designed and simulated a **CMOS inverter** using **GPDK 180nm**, performing detailed **DC and transient analysis** to optimize switching characteristics and minimize **rise/fall times**.

Twitter-Based Sentiment Analysis: Russia-Ukraine Conflict

Course Project-DMS672: Data Mining and Knowledge Discovery | Mentor: Dr. Faiz Hamid | IIT Kanpur (Jul'25)

- Analyzed **0.7M tweets** in 64 languages using **RoBERTa** and implemented **Support Vector Classification (SVC)** to achieve **86.7% accuracy** in detecting public sentiment and emotional triggers like **anger** and **joy**.

Interpretable Spatial-Temporal Video Transformer for Deepfake Detection

Course Project-EE656A: AI, ML, Deep Learning & its Applications | Mentor: Prof. Nishchal K. Verma | IIT Kanpur (Aug'25)

- Developed a Deepfake detector using **Neural Networks** and **Attention-based modules**, achieving **90% accuracy** on the FaceForensics++ dataset.

CERTIFICATIONS

- Completed **Udemy** course on **VSD - Static Timing Analysis - I** with **Certificate of Completion**.

TECHNICAL SKILLS

- Programming:** Python, Verilog HDL, C, Linux (Bash scripting)
- Hardware / EDA Tools:** Xilinx Vivado, Cadence Virtuoso, ModelSim
- Software:** MATLAB, Simulink, GitHub, VS Code

RELEVANT COURSES

Computer Architecture Electronic Devices and Circuits	Microprocessors & Interfacing Data Mining	Digital and Linear Electronic Circuits AI, ML, Deep Learning and Its Applications
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AREAS OF INTEREST & HOBBIES

- Areas of Interest:** Hardware Verification | Computer Architecture | RTL Design | Digital IC Design | FSM | STA
- Hobbies:** Singing | Astronomy | Sketching

POSITIONS OF RESPONSIBILITY

- Social Media Marketing Manager:** Electrical Engineering Association, EE Department, IIT Kanpur. (Aug'25-Present)
- Teaching Assistant:** Computer Vision and Deep Learning (EE655). (Jan'26-Present)
- Overall Management Teaching Assistant:** Introduction To Electronics Lab (ESC201). (July'25-Dec'25)
- Teaching Assistant:** Introduction To Electrical Engineering Lab (ESO 203). (Jan'25-Jun'25)